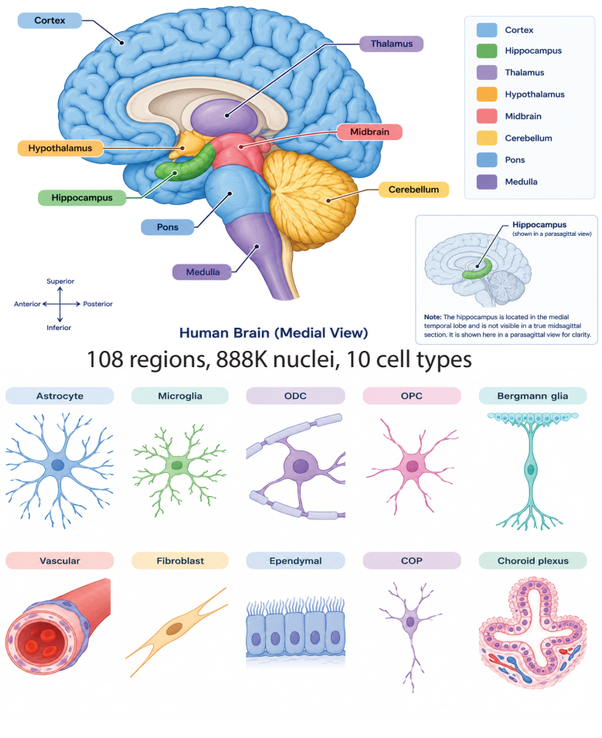
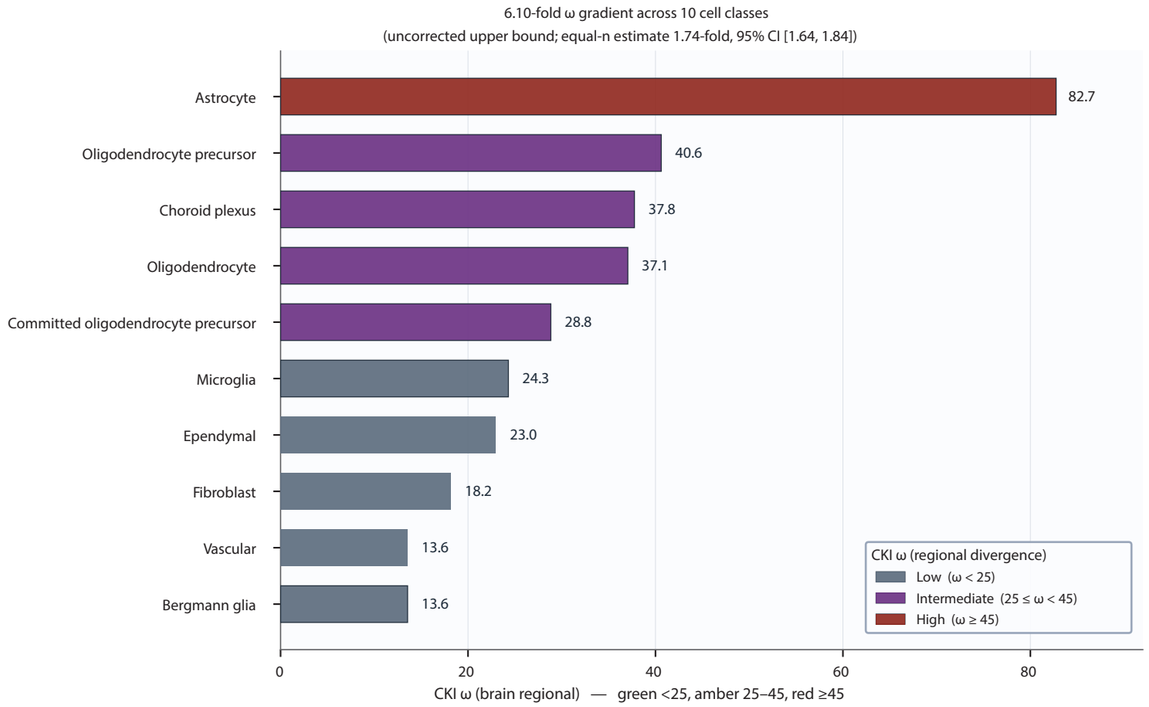
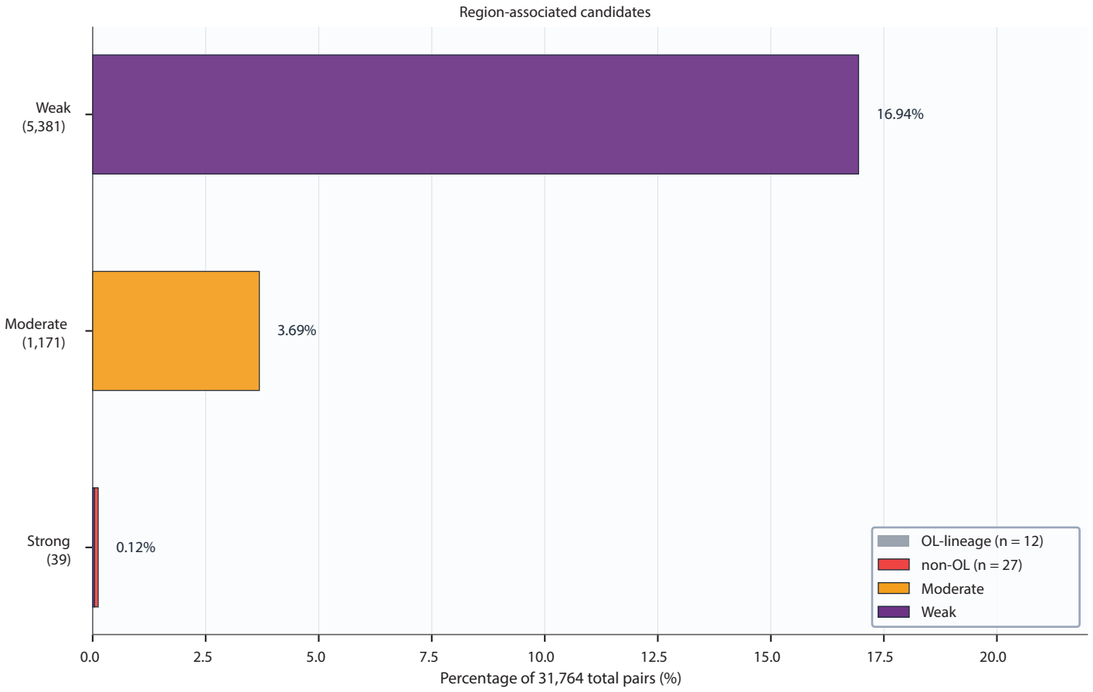
(a)


Cortex
Thalamus
Hypothalamus
Midbrain
Hippocampus
Cerebellum
Pons
Medulla
Cortex
Hippocampus
Thalamus
Hypothalamus
Midbrain
Cerebellum
Pons
Medulla
Hippocampus
(shown in a parasagittal view)
Note: The hippocampus is located in the medial temporal lobe and is not visible in a true midsagittal section. It is shown here in a parasagittal view for clarity.
Superior
Anterior
Posterior
Inferior
Human Brain (Medial View)
108 regions, 888K nuclei, 10 cell types
Astrocyte
Microglia
ODC
OPC
Bergmann glia
Vascular
Fibroblast
Ependymal
COP
Choroid plexus

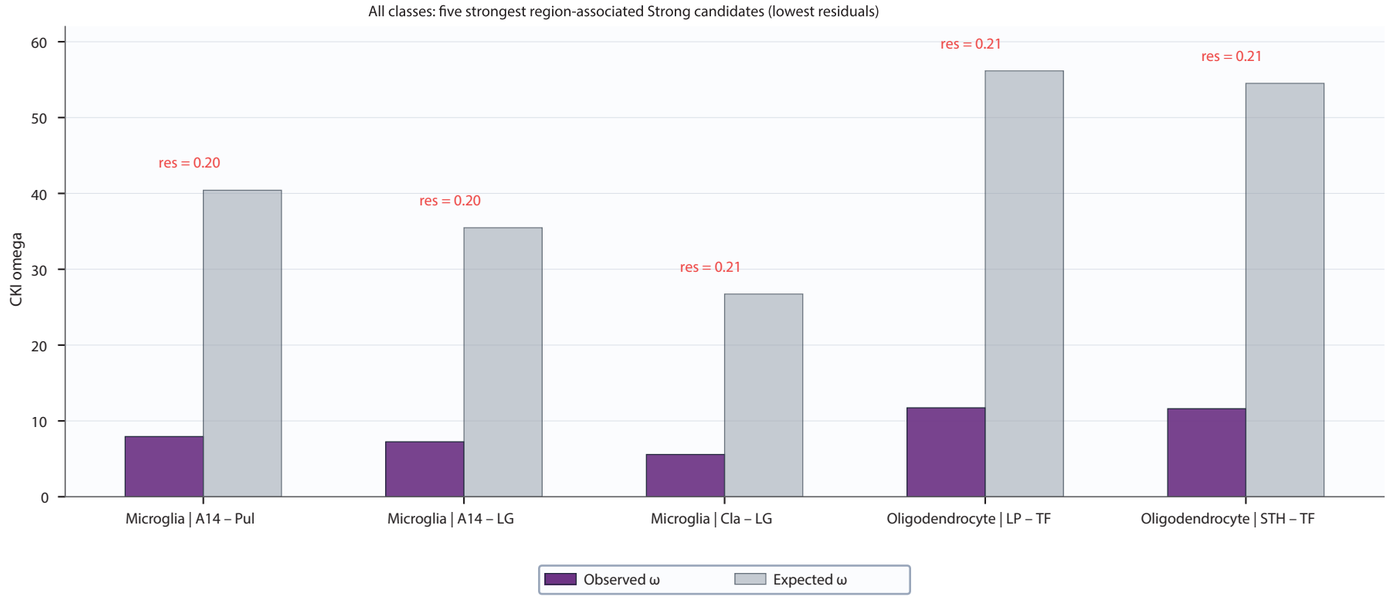
(b)


6.10-fold ω gradient across 10 cell classes
(uncorrected upper bound; equal-n estimate 1.74-fold, 95% CI [1.64, 1.84])
Astrocyte 82.7
Oligodendrocyte precursor 40.6
Choroid plexus 37.8
Oligodendrocyte 37.1
Committed oligodendrocyte precursor 28.8
Microglia 24.3
Ependymal 23.0
Fibroblast 18.2
Vascular 13.6
Bergmann glia 13.6
CKI ω (regional divergence)
Low (ω < 25)
Intermediate (25 ≤ ω < 45)
High (ω ≥ 45)
0
20
40
60
80
CKI ω (brain regional) — green <25, amber 25–45, red ≥45

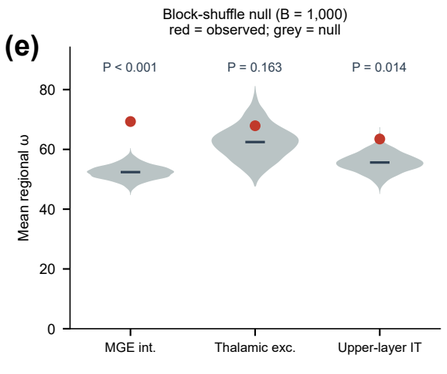
(c)


Region-associated candidates
Weak (5,381) 16.94%
Moderate (1,171) 3.69%
Strong (39) 0.12%
OL-lineage (n = 12)
non-OL (n = 27)
Moderate
Weak
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Percentage of 31,764 total pairs (%)

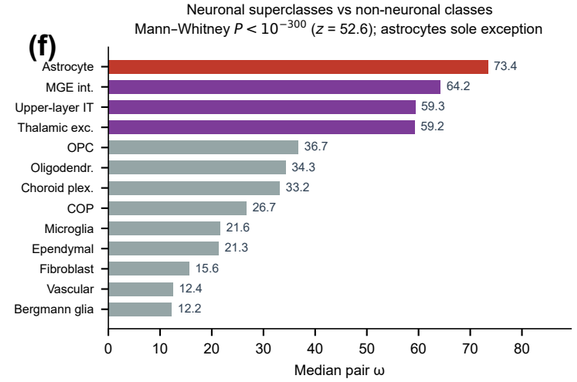
(d)


All classes: five strongest region-associated Strong candidates (lowest residuals)
CKI omega
res = 0.20
res = 0.20
res = 0.21
res = 0.21
res = 0.21
Microglia | A14 – Pul
Microglia | A14 – LG
Microglia | Cla – LG
Oligodendrocyte | LP – TF
Oligodendrocyte | STH – TF
Observed ω
Expected ω

(e)


Block-shuffle null (B = 1,000)
red = observed; grey = null
P < 0.001
P = 0.163
P = 0.014
Mean regional ω
MGE int.
Thalamic exc.
Upper-layer IT

(f)


Neuronal superclasses vs non-neuronal classes
Mann–Whitney P < 10−300 (z = 52.6); astrocytes sole exception
Astrocyte 73.4
MGE int. 64.2
Upper-layer IT 59.3
Thalamic exc. 59.2
OPC 36.7
Oligodendr. 34.3
Choroid plex. 33.2
COP 26.7
Microglia 21.6
Ependymal 21.3
Fibroblast 15.6
Vascular 12.4
Bergmann glia 12.2
Median pair ω